



Anode-free non-explosive EV battery

POSTECH research team led by Professor Soojin Park and Ph.D. candidate Sungjin Cho (Department of Chemistry) in collaboration with Professor Dong-Hwa Seo and Dr Dong Yeon Kim (School of Energy and Chemical Engineering) at Ulsan Institute of Science and Technology (UNIST) have developed an anode-free battery in a commonly used carbonate based liquid electrolyte by adding an ion conductive substrate. The substrate forms an anode protective layer and assists in minimising the bulk expansion of the anode. It is predicted that this battery will accelerate the commercialisation of non-explosive batteries since it maintains a high capacity for longer periods.

Motion detection technology that doesn't require sensors

Signify has introduced SpaceSense, a motion detection technology that doesn't require any sensor to be installed. This innovative technology uses Wi-Fi signals that are already present in the room to detect motion. This elimi-



Motion detection technology example (Credit: <https://www.signify.com>)

nates the need for dedicated sensors and batteries. The device measures small deviations in signal strength due to disturbance caused in the Wi-Fi signals when the people move around the room. The detection is omnidirectional. The sensitivity of the system can be adjusted to avoid any deviations from occurring. The system is accompanied by multiple other accessories which are optional for users, such as the Smart Button. It allows users to control lights with just one click of a button. This button as well is Wi-Fi operated. This technology can be used for a wide range of applications, may it be domestic or industrial.

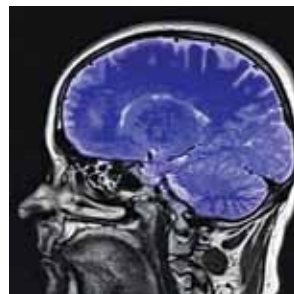
Simple, high-performance batteries made possible

Researchers at Colorado State University and the National Renewable Energy Laboratory have developed a framework consisting of AI tool AlphaZero coupled with a fast machine learning-derived model, made up of two graph neural networks trained on almost 100,000 quantum chemistry simulations. The first graph was trained to predict oxidation and reduction potentials. The second predicts the den-

sity of electrons and the local 3D environment. Researchers pose molecule optimisation as a tree search, where they build molecules by iterating components to add up into a growing structure. The advantage of this approach is that they prune off large branches of the search space where molecules start to show substructures that are unrealistic. On testing it was noted that molecules found could lead to simpler, high-performance batteries without requiring the use of transition metals.

Holographic microscope measures light's amplitude and phase

Researchers from Korea have invented the holographic microscope that could eliminate multiple scattering and simultaneously measure the amplitude and phase of light. This reflected light would provide the data required to observe the features located relatively deep within the brain tissues. The device was able to correct the wavefront distortion even at a depth that was previously impossible using existing technology. The team managed to quantitatively analyse the interaction between light and matter, which enabled them to further improve their previous microscope. This new microscope could focus more than 80 times of light energy on the neural fibres than before, while selectively removing unnecessary signals.



High-resolution image of the brain's neural network (Credit: www.ibs.re.kr)

Robotic cap in drug capsule can deliver drugs directly to gut

Researchers at MIT have developed a new drug capsule that consists of a robotic cap that allows the delivery of drugs to the gut without the need for injections. This unique drug capsule consists of a robotic cap that spins and tunnels through the mucus barrier when it reaches the small intestine. This mechanism allows the drugs carried by the capsule to pass into cells lining the intestine. The researchers showed how they could implement this approach to deliver insulin as well as vancomycin, an antibiotic peptide that currently has to be injected. The team envisions utilising this drug capsule to target the stomach or colon by changing the pH at which the gelatin coating dissolves.



Robot capsule for drug delivery (Credit: Traverso Lab/MIT and BWH)